Quantum Information

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Remark 1. *In this note all Hilbert spaces are complex and finite dimensional unless otherwise stated.*

1 Teleportation and dense coding

Let us assume that Alice and Bob share a maximally entangled state. Assume Alice has an additional other, unknown, state $|\Psi\rangle$. Quantum state teleportation is a protocol where Alice transfers the state $|\Psi\rangle$ to Bob using only two bits of classical communication. Even though it is required that at some point in time there is a qubit communication between Alice and Bob, one half of the maximally entangled state, this communication can happen Alice even gets the state $|\Psi\rangle$[1], thus it carries no information about $|\Psi\rangle$.

To describe the protocol, let us define the Bell basis. Let $\Omega \in \mathbb{C}^2 \otimes \mathbb{C}^2$ be the maximally entangled state, $\Omega = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$. Let us define $|\Omega_{\alpha\beta}\rangle$ as $|\Omega_{\alpha\beta}\rangle = (\mathrm{Id} \otimes X^\alpha Z^\beta)|\Omega\rangle$, $\alpha, \beta \in \{0, 1\}$. Explicitly,

$$
\begin{aligned}
|\Omega_{00}\rangle &= (\mathrm{Id} \otimes \mathrm{Id})|\Omega\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle, \\
|\Omega_{01}\rangle &= (\mathrm{Id} \otimes Z)|\Omega\rangle = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle, \\
|\Omega_{10}\rangle &= (\mathrm{Id} \otimes X)|\Omega\rangle = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle, \\
|\Omega_{11}\rangle &= (\mathrm{Id} \otimes XZ)|\Omega\rangle = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle.
\end{aligned}
$$

At the beginning of the protocol, Alice and Bob have the state $|\Psi\rangle \otimes |\Omega\rangle \in \mathcal{H}_{A'} \otimes \mathcal{H}_A \otimes \mathcal{H}_B$; in Alice's lab the qubits A' and A, in Bob's lab the qubit B. The protocol consists of the following steps:

- Alice measures the qubits A and A' in the Bell basis, and obtains outcome $(\alpha, \beta) \in \{0, 1\}^2$.
- Alice sends the classical bits α and β to Bob
- Bob applies on the particle B the operator $x^\alpha Z^\beta$.

We will show that at the end of the protocol, Bob obtains the state $|\Psi\rangle$.

2 Schmidt decomposition

Definition 1 (Schmidt decomposition)**.** *Let $\mathcal{H}, \mathcal{K}$ be Hilbert spaces, $|\Psi\rangle \in \mathcal{H} \otimes \mathcal{K}$. A Schmidt decomposition of $|\Psi\rangle$ is a finite set $\{(\lambda_i, |l_i\rangle, |r_i\rangle)\}_{i=1}^n \subseteq \mathbb{C} \times \mathcal{H} \times \mathcal{K}$ such that*

$$
|\Psi\rangle = \sum_{i=1}^n \lambda_i \cdot |l_i\rangle \otimes |r_i\rangle,
$$

and such that

[1] and actually it might be Bob who sends a qubit to Alice

- $\lambda_i > 0 \ \forall i = 1, \ldots, n$,
- $\langle l_i | l_j \rangle = \delta_{ij}$ *for all* $i, j = 1, \ldots, n$, *and*
- $\langle r_i | r_j \rangle = \delta_{ij}$ *for all* $i, j = 1, \ldots, n$.

The vectors $\{|l_i\rangle\}_{i=1}^n$ and $\{(|r_i\rangle\}_{i=1}^n$ are linearly independent, and thus a Schmidt decomposition is a minimal rank decomposition. Not every minimal rank decomposition is a Schmidt decomposition, as there the tensor components are not required to be orthonormal.

Example 1. Consider the following states.

- $|\Omega\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$. This is a Schmidt decomposition of $|\Omega\rangle$.
- The same state admits another Schmidt decomposition: $|\Omega\rangle = \frac{1}{\sqrt{2}}|++\rangle + \frac{1}{\sqrt{2}}|--\rangle$. So the Schmidt decomposition is not unique. Note: the lack of uniqueness is due to the degeneracy of its Schmidt values.
- $|\Psi\rangle = \frac{1}{\sqrt{2}}|+0\rangle + \frac{1}{\sqrt{2}}|11\rangle$ is not a Schmidt decomposition, as $\langle +|1\rangle \neq 0$.
- $|\Phi\rangle = \frac{1}{\sqrt{2}}|+0\rangle + \frac{1}{\sqrt{2}}|-1\rangle$ is a Schmidt decomposition. The basis on the left and on the right is not necessarily the same (they, in general, are even bases of different Hilbert spaces).

Theorem 1. *Let* $\mathcal{H}, \mathcal{K}$ *be Hilbert spaces, and* $|\Psi\rangle \in \mathcal{H} \otimes \mathcal{K}$, $|\Psi\rangle \neq 0$. *Then* $|\Psi\rangle$ *has a Schmidt decomposition.*

Proof. Let $\rho := \mathrm{Tr}_{\mathcal{K}} |\Psi\rangle\langle\Psi|$. This operator is positive semidefinite, and thus we can consider an eigen decomposition

$$\rho = \sum_i \lambda_i^2 \cdot |l_i\rangle\langle l_i|, \tag{1}$$

with $\lambda_i \in \mathbb{R}$, $\lambda_i \geq 0$ for all $i = 1 \ldots \dim \mathcal{H}$. The vectors $\{|l_i\rangle\}_{i=1}^{\dim \mathcal{H}}$ form a basis of $\mathcal{H}$, we can thus write

$$|\Psi\rangle = \sum_i |l_i\rangle \otimes |\hat{r}_i\rangle,$$

for some $\{|\hat{r}_i\rangle\}_{i=1}^{\dim \mathcal{H}} \subseteq \mathcal{K}$. Expressing ρ with the help of this form of $|\Psi\rangle$, we obtain

$$\rho = \mathrm{Tr}_{\mathcal{K}} |\Psi\rangle\langle\Psi| = \sum_{i,j=1}^{\dim \mathcal{H}} |l_i\rangle\langle l_j| \cdot \mathrm{Tr}\{|\hat{r}_i\rangle\langle\hat{r}_j|\} = \sum_{i,j=1}^{\dim \mathcal{H}} |l_i\rangle\langle l_j| \cdot \langle\hat{r}_j|\hat{r}_i\rangle.$$

Comparing this to Eq. (1), we obtain $\langle\hat{r}_j|\hat{r}_i\rangle = \delta_{ij}\lambda_i^2$. For $\lambda_i \neq 0$ we can thus define $|r_i\rangle = \lambda_i^{-1}|\hat{r}_i\rangle$. These states are, by definition, orthonormal, and

$$|\Psi\rangle = \sum_{i:\lambda_i \neq 0} \lambda_i \cdot |l_i\rangle \otimes |r_i\rangle$$

is a Schmidt decomposition of $|\Psi\rangle$. □

Consider $|\Psi\rangle \in \mathcal{H} \otimes \mathcal{K}$, and set $\rho_{\mathcal{H}} := \mathrm{Tr}_{\mathcal{K}} |\Psi\rangle\langle\Psi|$ and $\rho_{\mathcal{K}} := \mathrm{Tr}_{\mathcal{H}} |\Psi\rangle\langle\Psi|$. These two matrices – the reduced densities of the state $|\Psi\rangle$ – have the same spectrum, except for the eigenvalue 0: they are both the squared Schmidt values of $|\Psi\rangle$, plus maybe additional zeros.

3 LOCC

The laws of physics are inherently local (gravity might be an interesting exception): that is, only physical systems that are close to each other tend to interact. In our description, this means that if the two components of a bipartite quantum system AB are "far from each other", as for example, a pair of atoms located in two different labs, then there are operations that are "easy" to do and operations that are "hard" to do. While the exact boundary between easy and hard operations depend on the actual physical system, we can assume that *local* operations are easy to do: unitaries that are of the form $U_A \otimes \mathrm{Id}_B$ or $\mathrm{Id}_A \otimes U_B$, and measurements of the form $\{M_i^A \otimes \mathrm{Id}_B\}_{i=1}^n$ and $\{\mathrm{Id} \otimes M_i^B\}_{i=1}^n$. Applying a sequence of "easy" operations is considered to be "easy" again.

LOCC operations (local operations assisted by classical communication) consist of these local operations, but where we assume in addition that the parties can communicate with each other (by classical means): they can share their measurement outcomes, and thus the subsequent unitaries and measurements of all parties might depend on the measurement outcomes. We have already seen such a protocol: teleportation.

In this section, we will try to understand when a state $|\Phi\rangle$ can be transformed into another state $|\Psi\rangle$ via LOCC operations; if it is possible, we write $|\Phi\rangle \xrightarrow{LOCC} |\Psi\rangle$. In this case $|\Phi\rangle$ is a better resource for carrying out any task where the parties are restricted by locality, but where they are allowed to communicate: for example, when they want to carry out an experiment on a collection of entangled particles where the particles are located in different labs.

Theorem 2 (Ky-Fan)**.** *Let $\mathcal{H}$ be a Hilbert space, $k \in \mathbb{N}$, and*

$$\mathcal{P}_k(\mathcal{H}) := \left\{ P \in \mathcal{B}(\mathcal{H}) \middle| P = P^\dagger = P^2,\ \mathrm{Tr}(P) = k \right\}.$$

Let $\rho \in \mathcal{B}(\mathcal{H})$ Hermitian, and

$$\rho = \sum_{i=1}^{\dim \mathcal{H}} \lambda_i |\Phi_i\rangle\langle\Phi_i|$$

be its eigen decomposition, with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{\dim \mathcal{H}}$. Then

$$\sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{\rho P\} = \sum_{i=1}^{k} \lambda_i.$$

In other words, given a Hermitian operator ρ, the maximum of $\mathrm{Tr}\{\rho P\}$, where P is a rank-k Hermitian projector, is the sum of the k largest eigenvalues of ρ.

Proof. Let $Q = \sum_i |\Phi_i\rangle\langle\Phi_i|$. Then $Q \in \mathcal{P}_k$ and thus

$$\sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{\rho P\} \geq \mathrm{Tr}\{Q\rho\} = \sum_{i=1}^{k} \langle\Phi_i|\rho|\Phi_i\rangle = \sum_{i=1}^{k} \lambda_i. \tag{2}$$

To obtain the opposite bound, use the spectral decomposition of ρ to write

$$\mathrm{Tr}\{\rho P\} = \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \,\mathrm{Tr}\left\{P|\Phi_i\rangle\langle\Phi_i|\right\} = \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \cdot \langle\Phi_i|P|\Phi_i\rangle = \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \cdot \omega_i,$$

where $\omega_i = \langle\Phi_i|P|\Phi_i\rangle$ satisfies $0 \leq \omega_i \leq 1$ and $\sum_{i=1}^k \omega_i = k$. We thus obtain that

$$\sup_{P \in \mathcal{P}_k} \mathrm{Tr}\{\rho P\} \leq \sup_{\substack{\omega \in [0,1]^{\dim \mathcal{H}} \\ \sum_i \omega_i = k}} \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \cdot \omega_i.$$

The maximum of the r.h.s. is reached at $\omega_1 = \cdots = \omega_k = 1$, and $\omega_{k+1} = \cdots = \omega_{\dim \mathcal{H}} = 0$, and its value is $\sum_{i=1}^k \lambda_i$. Therefore

$$\sup_{P \in \mathcal{P}_k} \operatorname{Tr}\{\rho P\} \leq \sup_{\substack{\omega \in [0,1]^{\dim \mathcal{H}} \\ \sum_i \omega_i = k}} \sum_{i=1}^{\dim \mathcal{H}} \lambda_i \cdot \omega_i = \sum_{i=1}^k \lambda_i.$$

This, together with Eq. (2) is the desired statement. □

Definition 2 (Majorization)**.** *Let $p, q \in \mathbb{R}^n$ be probability distributions. Let $p^\downarrow$ ($q^\downarrow$) be the vector obtained by listing the entries of p (q) in descending order. We say that p is majorized by q, and write $p \preccurlyeq q$, if for all $k = 1, \ldots, n$,*

$$\sum_{i=1}^k p_i^\downarrow \leq \sum_{i=1}^k q_i^\downarrow.$$

TODO: Here it is better to define Schmidt coefficient with zero in it.

Proposition 1. *Let $|\Psi\rangle, |\Phi\rangle \in \mathcal{H} \otimes \mathcal{K}$. Let the squared Schmidt coefficients of $|\Psi\rangle$ be p, and the squared Schmidt coefficients of $|\Phi\rangle$ be q. If $|\Psi\rangle \xrightarrow{LOCC} |\Psi\rangle$, then $q \preccurlyeq p$.*

Proof. Let $\rho = \operatorname{Tr}_{\mathcal{H}} |\Psi\rangle\langle\Psi|$ and $\eta = \operatorname{Tr}_{\mathcal{H}} |\Psi\rangle\langle\Psi|$. Then the eigenvalues of ρ are precisely the entries of p and the eigenvalues of η are the entries of q. Using MISSING, $|\Psi\rangle \xrightarrow{LOCC} |\Psi\rangle$ if and only if

$$\sqrt{r_i}|\Phi\rangle = M_i \otimes U_i |\Psi\rangle,$$

for a measurement $\{M_i\}_{i=1}^n$, a probability distribution $r \in \mathbb{R}^n$ and some unitaries $\{U_i\}_{i=1}^n$. Therefore

$$r_i \eta = U_i \rho_i U_i^\dagger,$$

where $\rho_i = \operatorname{Tr}_{\mathcal{H}}\{(M_i^\dagger M_i \otimes \mathrm{Id})|\Psi\rangle\langle\Psi|\}$, and thus

$$\sup_{P \in \mathcal{P}_k} \operatorname{Tr}\{P\eta\} = \sup_{P \in \mathcal{P}_k} \sum_i \operatorname{Tr}\{r_i P \eta\} = \sup_{P \in \mathcal{P}_k} \sum_i \operatorname{Tr}\{P U_i \rho_i U_i^\dagger\} \leq \sum_i \sup_{P \in \mathcal{P}_k} \operatorname{Tr}\{U_i^\dagger P U_i \rho\} = \sum_i \sup_{P \in \mathcal{P}_k} \operatorname{Tr}\{P \rho_i\} = \sup_{P \in \mathcal{P}_k} \operatorname{Tr}\{P\rho\}.$$

Using now Theorem 2, we obtain that $p \preccurlyeq q$. □

Let us try to understand now a *sufficient* criterion for $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$.

Example 2. Let $|\Psi\rangle = \sqrt{1/2}|00\rangle + \sqrt{1/2}|11\rangle$ and $|\Phi\rangle = \sqrt{2/3}|00\rangle + \sqrt{1/3}|11\rangle$. Notice that

$$M_0 = \begin{pmatrix} \sqrt{2/3} & 0 \\ 0 & \sqrt{1/3} \end{pmatrix},$$

is a matrix such that $(M_0 \otimes \mathrm{Id})|\Psi\rangle = |\Phi\rangle$. Moreover, we can complete it to a measurement, for example, with the matrix

$$M_1 = \begin{pmatrix} \sqrt{1/3} & 0 \\ 0 & \sqrt{2/3} \end{pmatrix},$$

i.e., $M_0^\dagger M_0 + M_1^\dagger M_1 = \mathrm{Id}$. Notice that $(M_1 \otimes \mathrm{Id})|\Psi\rangle = X \otimes X |\Phi\rangle$. Therefore the measurement $N_i = X^i M_i$ ($i = 1, 2$) is such that

$$M_i \otimes X^i |\Psi\rangle = \sqrt{p_i}|\Phi\rangle,$$

for some probability distribution p_i, i.e., $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$.

The essence of this example is that the squared Schmidt coefficients of the states are related to each other as

$$\begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} = \frac{1}{2} \cdot (\mathrm{Id} + X) \cdot \begin{pmatrix} 1/3 \\ 2/3 \end{pmatrix}.$$

This construction can be easily generalized for the case when the squared Schmidt coefficients of $|\Psi\rangle$ can be obtained by mixing permutations of the squared Schmidt coefficients of $|\Phi\rangle$. Before we show how to obtain the LOCC operations for this generalized case, notice that this constraint is equivalent with majorization:

Lemma 1. *Let $p, q \in \mathbb{R}^n$ be probability distributions. If $p \preccurlyeq q$, then there is a probability distribution $r \in \mathbb{R}^{n!}$ such that*

$$p = \sum_{\pi \in S_n} r_\pi \pi q,$$

where $S_n \subseteq \mathcal{B}(\mathbb{R}^n)$ is the set of $n \times n$ permutation matrices.

Proof. Exercise. □

Again, number of non-zero Schmidt coeffs might be different!

Proposition 2. *Let $|\Psi\rangle, |\Phi\rangle \in \mathcal{H} \otimes \mathcal{K}$. Let the squared Schmidt coefficients of $|\Psi\rangle$ be p, and the squared Schmidt coefficients of $|\Phi\rangle$ be q. If $p \preccurlyeq q$, then $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$.*

Proof. Using Lemma 1, we obtain that there is a probability distribution r such that $p = \sum_{\pi \in S_n} r_\pi \pi q$, where S_n is the group of $n \times n$ permutation matrices. Let the Schmidt decomposition of the states be

$$|\Psi\rangle = \sum_{i=1}^{n} p_i \cdot |l_i\rangle \otimes |r_i\rangle$$
$$|\Phi\rangle = \sum_{i=1}^{n} q_i \cdot |\hat{l}_i\rangle \otimes |\hat{r}_i\rangle,$$

and let us define operators M_π as

$$M_\pi = \sum_i \left(r_\pi \frac{q_{\pi(i)}}{p_i} \right)^{1/2} |\hat{l}_{\pi(i)}\rangle\langle l_i|.$$

These operators form a POVM as

$$\sum_\pi M_\pi^\dagger M_\pi = \sum_{ij} \sum_\pi r_\pi \frac{q_{\pi(i)}}{p_i} |l_i\rangle\langle \hat{l}_{\pi(i)} | \hat{l}_{\pi(j)}\rangle\langle l_j| = \sum_i \sum_\pi r_\pi \frac{q_{\pi(i)}}{p_i} |l_i\rangle\langle l_i| = \sum_i |l_i\rangle\langle l_i| = \mathrm{Id}.$$

Note as well that

$$(M_\pi \otimes \mathrm{Id})|\Psi\rangle = \sum_i \sqrt{r_\pi q_{\pi(i)}} |\hat{l}_{\pi(i)}\rangle \otimes |r_i\rangle,$$

and thus, setting $V_\pi = \sum_i |r_{\pi(i)}\rangle\langle r_i|$, V_π is a unitary and

$$(M_\pi \otimes V_\pi)|\Psi\rangle = \sum_i \sqrt{r_\pi q_{\pi(i)}} |\hat{l}_{\pi(i)}\rangle \otimes |r_{\pi(i)}\rangle = \sqrt{r_\pi} |\Phi\rangle,$$

i.e., we have shown that $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$. □

We have thus seen in Proposition 1 and Proposition 2 that

Theorem 3. *Let $|\Psi\rangle, |\Phi\rangle \in \mathcal{H} \otimes \mathcal{K}$. Let the squared Schmidt coefficients of $|\Psi\rangle$ be p, and the squared Schmidt coefficients of $|\Phi\rangle$ be q. Then $|\Psi\rangle \xrightarrow{LOCC} |\Phi\rangle$ iff $p \preccurlyeq q$.*